Quantum Algorithm for Computing the Period Lattice of an Infrastructure

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Joint work

- *Quantum Algorithms for One-Dimensional Infrastructures*

  with Pradeep Sarvepalli (University of British Columbia)

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- *Quantum Algorithm for Computing the Period Lattice of an Infrastructure*

  with Felix Fontein (University of Zürich)

  preprint to appear soon on the arXiv

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Motivation and overview

- infrastructures are group-like algebraic objects
- they appear in arithmetic geometry in the study of computational problems related to algebraic number fields and function fields with finite constant fields
- arithmetic geometry provides a unified treatment of global fields
- the most prominent computational tasks are computing
  - the period lattice of the infrastructure and
  - generalized discrete logarithms
- both problems are not known to have efficient classical algorithms

QA for problems in algebraic geometry

- there is an elegant and efficient quantum algorithm for infrastructures arising from function fields (of curves over finite fields)
- such infrastructures are discrete and embed into finite abelian groups (divisor class groups of degree zero)
- there are efficient (classical) methods to do arithmetics in these groups
- $\Rightarrow$ we can simply use the quantum algorithm for the abelian HSP to compute the period lattice of discrete infrastructures

QA for problems in algebraic geometry

- the quantum algorithm for the abelian HSP can also be used to efficiently
  - compute generalized DLOGs in discrete infrastructures
  - compute the whole divisor group class and the ideal class group
  - solve the principal ideal problem
  - compute the Zeta function
- $\Rightarrow$ we focus our attention on non-discrete infrastructures
- such infrastructures arise from number fields

One-dimensional pseudo-periodic states

- let $\Lambda = R\mathbb{Z}$ with $R \in \mathbb{R}_{>0}$
- a quantum state in $\mathbb{C}^{qN}$ of the form

$$|\psi\rangle \propto \sum_{\lambda \in \Lambda \cap [0,qN)} |v + N\lambda + \xi_\lambda\rangle \quad \text{with } \xi_\lambda \in (-1,1)$$

  is called a pseudo-periodic with period R (with period lattice Λ)
- $v \in \{0, 1, \ldots, \lfloor NR \rfloor\}$ is random
- there are approximately $\frac{q}{\det(\Lambda)} = \frac{q}{R}$ elements in $|\psi\rangle$
- task: find a rational number $\hat{R}$ such that $|\hat{R} - R| \leq \gamma$

QA for estimating the period

- INPUT:
  - two positive integer q and N with $q > 2R$ and $N > \frac{2}{R}$
  - two pseudo-periodic states in $\mathbb{C}^{qN}$ with period R
- embed each state into a register $\mathbb{C}^{2qN}$
- apply the QFT of size $2qN$ to each state
- measure both registers
- denote the outcomes by w_1 and w_2

Sampling approximations of Λ^*

- let $\Lambda^* = \frac{1}{R}\mathbb{Z}$ be the dual lattice of $\Lambda = R\mathbb{Z}$
- let $\mathcal{Y}$ be the set of $w \in \{0, 1, \ldots, 2qN\}$ satisfying
  - $w_i = [2q\lambda^*]$ and for some $\lambda^* \in \Lambda^*$ and
  - $0 \leq w_i \leq 2q(\kappa N)$ with $\kappa \in (0, 1)$
- for all $w \in \mathcal{Y}$, we have

$$\Pr(w) \geq \frac{1}{2qN} \cos^2\left(\pi(\tfrac{1}{4} + \tfrac{1}{4qN} + 2\kappa)\right) \frac{q}{R}$$

- the cardinality of $\mathcal{Y}$ is approximately $\frac{\kappa N}{\det(\Lambda^*)} = \kappa N R$

Approximate generating set for Λ^*

- Let a and b be two numbers chosen uniformly at random from $\{1, 2, \ldots, B\}$ for $B \geq 2$

$$\Pr\big(\gcd(a,b) = 1\big) \geq 1 - \sum_{p \text{ prime}} \frac{1}{p^2} > \frac{1}{2}$$

- for half of the pairs $(w_1, w_2) \in \mathcal{Y} \times \mathcal{Y}$, the corresponding λ_i^* generate Λ^*
- $\Rightarrow$ the pair $(\frac{w_1}{2q}, \frac{w_2}{2q})$ is an approximate generating set of Λ^* with probability at least

$$\frac{\kappa^2}{8} \cos^4\big(\pi(\tfrac{1}{4} + \tfrac{1}{4qN} + 2\kappa)\big) \geq 10^{-5} \quad \text{for } \kappa = \tfrac{1}{16}$$

Approximate basis for Λ

- we can recover an approximate basis of Λ^* from the approximate generating set formed by $\frac{w_1}{2q}$ and $\frac{w_1}{2q}$
- we can recover an approximate basis of Λ^*, i.e., $\hat{R}$ by taking the inverse of the approximate basis of Λ^*

Success probability of the QA

- our algorithm yields the approximation $\hat{R}$ with probability at least 10^{-5}
- Hallgren's analysis yields the significantly worse bound $\Omega(1/\log^4(M))$ where M is an upper bound on RN and works only if $\log(M) \geq 256$

$$\frac{1}{256^4} = \frac{1}{4\,294\,967\,296} < \frac{1}{4 \cdot 10^9}$$

$\Rightarrow$ the guaranteed expected running time would be at least 50 days if it took 1 ms to execute the quantum algorithm once

One-dimensional infrastructures

- the problem of computing the period lattice of a one-dimensional infrastructure

$$\Downarrow$$

  the problem of estimating the period of a one-dimensional periodic quantum state

- one-dimensional infrastructures arise from number fields of unit rank one
- quadratic number fields $\mathbb{Q}(\sqrt{d})$ with d positive and squarefree are only one type of number fields of unit rank one

Improvements

- quadratic number fields yield one-dimensional IS
  $\Rightarrow$ our algorithms can be used to solve Pell's equation and the principal ideal problem
- in this sense, they generalize Hallgren's QA
- our improvements and extensions:
  - give a conceptually simpler presentation
  - obtain an improved technique of analyzing Fourier sampling of one-dimensional pseudo-periodic states
  - find algorithms with lower complexity and constant success probability

One-dimensional infrastructure

- a one-dimensional infrastructure of circumference R is a pair (X, d) where
  - $X = \{x_0, x_1, \ldots, x_{m-1}\}$ is a finite set and
  - $d : X \to \mathbb{R}/R\mathbb{Z}$ is injective
- the elements of X are ordered such that

$$0 = d(x_0) < d(x_1) < \ldots < d(x_{m-1}) < R$$

  we use representatives in $[0, R)$ to order the $d(x_i)$

- imagine that we place the elements around a circle of circumference R starting at the top most point and moving clockwise

Baby step

- the baby step $\mathrm{bs} : X \to X$ is defined as

$$\mathrm{bs}(x_i) = \begin{cases} x_{i+1} & 0 \leq i < m-1 \\ x_0 & i = m-1 \end{cases}$$

  it gives you the next element on the circle

- the relative distance function $\Delta_{\mathrm{bs}} : X \to [0, R)$ is such that

$$d(\mathrm{bs}(x)) = d(x) + \Delta_{\mathrm{bs}}(x)$$

  holds for all $x \in X$

  it tells you how far you have to travel along the circle to get to the next element

Giant step

- for $x, y \in X$ consider the set

$$S_{x,y} = \{s \in [0, R) \mid d(x) + d(y) + s \in d(X)\}$$

- the giant step $\mathrm{gs} : X \times X \to X$ is defined as

$$\mathrm{gs}(x, y) = z$$

  such that

$$d(z) = d(x) + d(y) + \min S_{x,y}$$

- the relative distance function $\Delta_{\mathrm{gs}} : X \times X \to [0, R)$ is defined as

$$\Delta_{\mathrm{gs}}(x, y) = \min S_{x,y}$$

Computational problems

- estimate the circumference R
- given $x \in X$, estimate $d(x)$
- these problems are computationally hard for classical computers
- the first generalizes the problem of computing the order of a (black-box) cyclic group
- the second generalizes the problem of determining discrete logarithms in a (black-box) cyclic group

Finite cyclic groups

- suppose $G = \langle g \rangle$ is a finite cyclic group of order R
- $\Rightarrow$ we can form an infrastructure out of G as follows:
  - $X = G$
  - $d(h) = \log_g(h)$
  - the baby step corresponds to multiplication by g
  - the giant step corresponds to the multiplication of two elements in G
  - Δ_{bs} always takes on the value 1
  - Δ_{gs} always takes on the value 0

Group arithmetic with f-reps

- a pair $(x, t) \in X \times \mathbb{R}$ is called an f-representation if

$$0 \leq t < \Delta_{\mathrm{bs}}(x)$$

- denote the set of all f-representations by $\mathrm{Rep}(\mathcal{I})$
- $\mathrm{Rep}(\mathcal{I})$ can be endowed with a group structure such that

$$\mathrm{Rep}(\mathcal{I}) \cong \mathbb{R}/R\mathbb{Z}$$

- set

$$\Lambda = R\mathbb{Z}$$

HSP over $\mathbb{R}$

- we have a surjective group homomorphism

$$f : \mathbb{R} \to \mathrm{Rep}(\mathcal{I}) \quad \text{with} \quad \ker f = \Lambda$$

- $\Rightarrow$ intuition: determine the subgroup Λ of $\mathbb{R}$ hidden by f
- but there are two fundamental problems:
  - we can only evaluate at a finite number of points
  - we cannot compute f exactly due to finite precision since Δ_{bs} and Δ_{gs} can only be approximated
- $\Rightarrow$ we have to work with a finite number of carefully chosen evaluation points that avoid “forbidden intervals”

Computable version

- Let $h : \mathcal{V} \to X \times \mathcal{V}$ be defined as

$$h(v) = (x, \lfloor Nt \rfloor)$$

  - $\mathcal{V} = \{0, \ldots, qN - 1\}$
  - s is some random offset
  - $f(s + \frac{v}{N}) = (x, t)$

- we can compute $h(v) = (x, k)$ correctly for all $v \in \mathcal{V}$ with high probability

- for most $(x, k) \in h(\mathcal{V})$, the preimage has the special form

$$h^{-1}(x, k) = \{v + N\lambda + \xi_\lambda \,:\, \lambda \in \Lambda\} \subseteq \mathcal{V}, \quad \xi_j \in (-1, 1)$$

Reduction to pseudo-periodic states

- evaluate h in superposition over $\mathcal{V}$ and measure
- this process prepares a pseudo-periodic state in $\mathbb{C}^{qN}$

$$|\psi\rangle \propto \sum_{\lambda \in R\mathbb{Z} \cap [0, qN)} |v + N\lambda + \xi_\lambda\rangle \quad \text{with } \xi_\lambda \in (-1, 1)$$

- $\Rightarrow$ we can apply our quantum algorithm to obtain the estimate $\hat{R}$

Higher-dim. pseudo-periodic states

- let Λ be a full-rank lattice in $\mathbb{R}^n$
- a quantum state in $(\mathbb{C}^{qN})^{\otimes n}$ of the form

$$\sum_{\lambda \in \Lambda \cap [0,qN)^n} |v + N\lambda + \xi_\lambda\rangle, \quad \xi_\lambda \in (-1,1)^n$$

  is called pseudo-periodic state with period lattice Λ
- Task: find an approximate basis $\lambda_1', \dots, \lambda_n'$ of Λ
- $\lambda_1', \dots, \lambda_n'$ is called a γ-approximate basis iff
  - $\Lambda = \mathbb{Z}\lambda_1 + \ldots + \mathbb{Z}\lambda_n$
  - $\|\lambda_i - \lambda_i'\|_2 \leq \gamma$

Higher-dimensional infrastructures

- an n-dimensional infrastructure $\mathcal{I}$ is a finite set of distinguished points on an n-dimensional torus $\mathbb{R}^n/\Lambda$
- Λ is a lattice of full rank in $\mathbb{R}^n$
- to each distinguished point x, we assign a region on the torus, so that every point on the torus lies in exactly one such region
- $\Rightarrow$ every point y in the region of x can be represented by the difference $t := y - x$ together with x, i.e. as the pair (x, t)
- these tuples (x, t) are called the f-representations of the infrastructure

Group arithmetic with f-reps.

- denote the set of all f-representations by $\mathrm{Rep}(\mathcal{I})$
- $\mathrm{Rep}(\mathcal{I})$ can be endowed with a group structure such that

$$\mathrm{Rep}(\mathcal{I}) \cong \mathbb{R}^n/\Lambda$$

HSP over $\mathbb{R}^n$

- exactly as in the one-dimensional case, we have a surjective group homomorphism

$$f : \mathbb{R}^n \to \mathrm{Rep}(\mathcal{I}) \quad \text{with} \quad \ker f = \Lambda$$

- $\Rightarrow$ intuition: determine the subgroup Λ of $\mathbb{R}^n$ hidden by f
- but there are two fundamental problems:
  - we can only evaluate at a finite number of points
  - we cannot compute f exactly because we cannot can only approximate f-representations
- $\Rightarrow$ we have to work with a finite number of carefully chosen evaluation points that avoid “forbidden regions”

Forbidden regions

Reduction to pseudo-periodic states

- evaluate h in superposition over $\mathcal{V} = \{0, \ldots, qN - 1\}^n$ and measure
- with high probability, we obtain the pseudo-periodic state

$$\sum_{\lambda \in \Lambda \cap [0, qN)^n} |v + N\lambda + \xi_\lambda\rangle, \quad \xi_\lambda \in (-1, 1)^n$$

QA for estimating the period lattice

- INPUT:
  - two positive integers q and N that are sufficiently large
  - $2n+1$ pseudo-periodic states in $(\mathbb{C}^{qN})^{\otimes n}$ with period lattice Λ
- embed each state into a register $(\mathbb{C}^{2nqN})^{\otimes n}$
- apply the multidimensional QFT to each state
- measure
- denote the outcomes by $w_1, \ldots, w_{2n+1}$

Sampling approximations of Λ^*

- let Λ^* be the dual lattice of Λ
- let $\mathcal{Y}$ be the set of $w \in \{0, 1, \ldots, 2nqN - 1\}^n$ satisfying
  - $w_i = [2nq\lambda^*]$ and for some $\lambda^* \in \Lambda^*$ and
  - $\|w_i\|_\infty \leq 2nq(\kappa N)$ with $\kappa \in (0, 1)$
- for all $w \in \mathcal{Y}$, we have

$$\Pr(w) \geq \frac{1}{(2qnN)^n} \cos^2\left(\pi(\tfrac{1}{4} + \tfrac{1}{4qN} + 2\kappa)\right) \frac{q^n}{\det(\Lambda)}$$

- the cardinality of $\mathcal{Y}$ is approx. $\frac{(\kappa N)^n}{\det(\Lambda^*)} = (\kappa N)^n \det(\Lambda)$

Approximate generating set for Λ^*

- we sample n vectors of $\Lambda^* \cap [0, b)^n$
- with prob. at least $\frac{1}{4}$, they generate a full-rank sublattice Λ_0^*
- Λ^*/Λ_0^* is a finite group generated by n elements
- we sample $n+1$ vectors of $\Lambda^* \cap [0, b_0)^n$ with $b_0 > b$
- this induces an almost uniform distribution of Λ^*/Λ_0^*
- with prob. at least $\frac{1}{4} - \hat{\zeta} > 0.184$ the new and old samples generate Λ^*

Approximate basis for Λ

- we can recover an approximate basis of Λ^* from the approximate generating set formed by

$$\frac{w_1}{2nq}, \ldots, \frac{w_{2n+1}}{2nq}$$

- we can recover an approximate basis of $\Lambda^*$from the approximate basis of Λ^*
- the algorithm succeeds with probability at least

$$\frac{1}{4}\Big(\frac{1}{4} - \hat{\zeta}\Big)\left(\frac{c\kappa^n}{(2n)^n}\right)^{2n+1}$$

where $c = \cos^2\left(\pi(\frac{1}{4} + \frac{1}{4qN} + 2\kappa)\right)$ and $\kappa = \frac{1}{16n}$

Improvements

- we present a rigorous and complete proof that the running time is polynomial in the determinant of the period lattice Λ

- we give an explicit lower bound on the success probability

  HALLGREN and SCHMIDT AND VOLLMER did not give any lower bound

  SCHMIDT presented a lower bound in his dissertation which is worse than our bound

- message: it is very important to improve the lower bound !

  we are sure that this can be done !

Thank you!!!